

ZeroGraph: Replacing the Global Backward Graph with Materialized Supervision

Omar Ramadan
Independent Researcher

Abstract—End-to-end backpropagation turns model depth into a synchronous information dependency: an upstream update cannot be computed until downstream credit arrives. ZEROGRAPH changes this systems contract by partitioning residual depth into independently optimized denoising jobs with disjoint parameter ownership, no inter-block autograd edges, and zero strict-mode inter-block gradient traffic. Independence alone is insufficient, however, because useful intermediate semantics must still compose. A preregistered GPT-2 Large experiment on four disjoint L4 GPUs reduced peak allocated HBM by $2.41\times$ and shortened the measured local critical path by $6.30\times$ relative to matched DDP, yet failed context-use and generation-retention gates: every local loss improved while all measured low-noise context effects were negative. We therefore replace the missing online dependency with cached, noise-conditioned teacher boundary targets. In a revision-pinned 1.100B TinyLlama/UltraChat conversion, this materialized-supervision regime used $4.62\times$ less peak HBM and had a $4.90\times$ shorter local critical path than matched DDP. Assembled held-out cross entropy was 1.3096, versus 1.2791 for DDP and 1.3256 for the teacher; a confirmation seed passed every preregistered gate. Cache construction and assembly are excluded from the local timing and remain explicit in our accounting. A new six-arm, 2.10M-token comparison against four-way FSDP2 isolates the mechanism: cached $\sigma = 0$ boundary regression passed every generation gate at CE 1.3059, whereas cached and online noise-conditioned variants were bitwise identical but both collapsed in free-running generation despite CE 1.3173. The successful boundary-regression arm’s cache-inclusive first use was $4.079\times$ shorter than FSDP2 in this allocation, while peak HBM was essentially equal. A subsequent nine-phase Qwen2.5 campaign adds an objective-matched cached/online comparison, ZeRO-3, $B \in \{2, 4, 8\}$, boundary-shift sweeps, exact failure/restart, three seeds, and a real 3.086B replication. Cached target acquisition was faster after materialization, but did not break even by $K = 16$; the 3B assembled model passed every quality gate at CE 1.3002 with zero inter-block gradient bytes. Thus materialized trajectories create independently schedulable depth-wise jobs, while neither a universal end-to-end speedup nor a benefit from the tested language diffusion objective is established.

Index Terms—blockwise learning, distributed training, local objectives, materialized supervision, knowledge distillation, memory-efficient training

I. INTRODUCTION

Large-model systems usually optimize the mechanics of a globally connected backward pass. Activation checkpointing trades recomputation for activation storage [1]; ZeRO and FSDP shard parameters, gradients, and optimizer state [2], [3]; tensor and pipeline parallelism distribute operators and depth [4], [5]. These methods change where state lives and when computation executes, but upstream credit still depends on information produced downstream.

For ordinary reverse-mode differentiation, partition b receives a cotangent $\lambda_b = \partial\mathcal{L}/\partial h_b$ from the suffix. The cotangent may be stored, recomputed, communicated, or losslessly encoded, but it cannot be discarded while preserving the ordinary gradient for every downstream computation. EXACTGRAPH, our objective-preserving control, rematerializes partitions but retains this dependency.

ZEROGRAPH takes the other branch of the design space: it changes the learning signal so each partition can optimize without a downstream gradient. The DiffusionBlocks construction assigns residual partitions denoising objectives over noise intervals [6]. The systems appeal is immediate: one backward graph and optimizer per block, zero strict-mode inter-block gradient traffic, and one independently schedulable job per partition. The scientific obligation is harder: independently learned representations must still compose.

Our experiments isolate that obligation. Unanchored GPT-2 Large blocks reduce their local losses and denoise targets while failing to use causal context or retain language behavior. Removing the global backward signal removes not only synchronization but also information constraining intermediate semantics. The positive experiment supplies that information ahead of optimization through *materialized boundary supervision*. A frozen teacher is run once over a fixed stream; its boundary states are cached; and block workers subsequently train without the teacher body or neighboring trainable blocks:

$$\begin{array}{c} \text{synchronous backward cotangents} \\ \Downarrow \text{move cross-depth information in time} \Downarrow \\ \text{materialized boundary targets} \end{array} \quad (1)$$

This is conversion/distillation, not from-scratch independent pretraining. Teacher boundary matching itself is not claimed as novel. The systems contribution is to make the materialized trajectory the cross-depth information interface, changing *when* and *where* global information is delivered.

Our contributions are:

- 1) We state the boundary between objective-preserving rematerialization and depth-independent optimization: exact upstream credit requires downstream information, while independent jobs require a replacement signal.
- 2) We implement ZEROGRAPH as a PyTorch partition/compiler runtime with explicit ownership, independent local jobs, resumable authenticated artifacts, and zero strict-mode inter-block autograd edges.

- 3) We report a preregistered negative GPT-2 Large experiment showing that unanchored denoising can optimize locally while intermediate semantics drift.
- 4) We isolate that signal in a six-arm TinyLlama comparison against four-way FSDP2: cached $\sigma = 0$ boundary regression preserves coherent language, while adding the tested noise objective breaks free-running generation despite strong teacher-forced CE.
- 5) We demonstrate bitwise equivalence of cached and online supervision, composition with inner FSDP2, explicit cache break-even accounting, a 4.027B systems fixture, and a guarded 1.007B EXACTGRAPH certificate.

II. FROM BACKWARD DEPENDENCIES TO MATERIALIZED SUPERVISION

Let a residual model be partitioned into B contiguous maps

$$h_b = F_b(h_{b-1}; \theta_b), \quad b = 1, \dots, B, \quad (2)$$

with ordinary loss $\mathcal{L}(h_B, y)$.

A. The Objective-Preserving Boundary

Reverse-mode differentiation applies

$$g_b = \left(\frac{\partial F_b}{\partial \theta_b} \right)^T \lambda_b, \quad \lambda_{b-1} = \left(\frac{\partial F_b}{\partial h_{b-1}} \right)^T \lambda_b. \quad (3)$$

Rematerialization can limit live state to a boundary and the partition being recomputed, but it does not remove λ_b as an information dependency.

Proposition 1 (reverse-dependency boundary). Consider two nonconstant differentiable partitions followed by an arbitrary differentiable scalar loss. No algorithm updating the first partition from only its locally visible forward information, while receiving neither a downstream adjoint nor information sufficient to reconstruct it, can equal the ordinary gradient for every suffix and loss.

Proof. Hold the input, first partition, and all locally visible values fixed. Choose two downstream compositions producing distinct cotangents at the same boundary; opposite linear terminal losses after an identity suffix suffice. A local algorithm must emit the same update because its observations are identical, while the chain rule in (3) changes the gradient whenever the local Jacobian has a component in the differing-adjoint direction. At least one ordinary gradient cannot be reproduced. $\square$

The proposition marks a systems boundary rather than a new chain-rule result. Exact execution may store, recompute, compress, or reconstruct the cotangent; independent depth-wise optimization needs another learning signal.

B. Unanchored Local Denoising

For target representation y and context x , sample $z_\sigma = y + \sigma\epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. Following score-based denoising [7], [8], split the noise support into intervals $I_b = [\sigma_b, \sigma_{b-1}]$. Block b minimizes

$$\mathcal{L}_b(\theta_b) = \mathbb{E}[w(\sigma)\ell(D_b(x, z_\sigma, \sigma; \theta_b), y)], \quad \sigma \sim p_b, \quad (4)$$

TABLE I
PLACEMENT OF CROSS-DEPTH INFORMATION.

Regime	Objective	Online signal	Local jobs
Ordinary	global	λ_b	no
EXACTGRAPH	global	exact λ_b	no
ZEROGGRAPH unanchored	local	none	yes
ZEROGGRAPH materialized	teacher-local	cached h_b^T	yes

where p_b is the base distribution restricted to I_b . With frozen or block-private interfaces, $\mathcal{L}_b$ contains only θ_b :

$$\frac{\partial \mathcal{L}_b}{\partial \theta_c} = 0 \quad (c \neq b), \quad \nabla_{\theta}^2 \sum_b \mathcal{L}_b \text{ is block diagonal.} \quad (5)$$

This gives zero inter-block gradient traffic. It does not require independently learned boundary representations to remain mutually compatible.

C. Materialized Boundary Supervision

Let a frozen teacher define $h_0^T, \dots, h_B^T$ on a fixed stream. Boundary states are generated before local optimization and stored as targets. Worker b then minimizes

$$z_{b,\sigma}^T = h_b^T + \sigma\epsilon, \quad (6)$$

$$\mathcal{J}_b(\theta_b) = \mathbb{E}[w(\sigma)\ell(D_b(h_{b-1}^T, z_{b,\sigma}^T, \sigma; \theta_b), h_b^T)] + R_b(\theta_b), \quad (7)$$

where R_b contains declared local regularizers or task terms. The TinyLlama recipe includes a clean initialization anchor and final-block assistant cross entropy. Once the cache exists, a worker needs neither the teacher body nor a neighboring trainable block.

This regime moves global information; it does not erase it. It also introduces a distribution shift: training conditions on h_{b-1}^T , whereas assembled inference receives a learned predecessor state. We therefore measure assembled boundary MSE and context ablations. Robustness to deliberately perturbed or mixed teacher/assembled boundaries remains an open ablation.

III. COMPILER, RUNTIME, AND COST MODEL

The compiler captures an executable PyTorch model with `torch.export` [9], identifies guarded residual separators, assigns parameters, buffers, and aliases to explicit owners, and emits one loadable artifact per block. Shape-changing heads and shared interfaces remain explicit. Strict independence requires shared trainable interfaces to be frozen or block-private; synchronization is possible but invalidates the zero-inter-block-communication contract.

Teacher boundaries are bound to model and dataset revisions, sample order, shapes, and content digests. Workers consume cached targets without loading the teacher during timed updates. Checkpoints bind block identity, program digest, optimizer, step, and RNG state. Training, assembly, quality evaluation, and evidence verification are separate operations, preventing local loss from standing in for assembled quality.

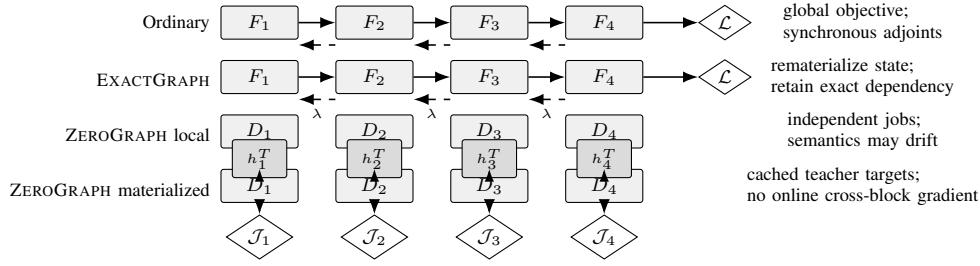


Fig. 1. Four execution semantics. Ordinary training and EXACTGRAPH retain the reverse dependency required by the global objective. Unanchored ZEROGRAPH removes it but can lose compatible semantics. Materialized ZEROGRAPH supplies a cached reference trajectory while keeping block optimization independent.

For block b with N_b parameters, effective state cost s_{state} , inner sharding degree D_b , and activation footprint A_b , active memory is approximately

$$M_b \approx \frac{s_{\text{state}} N_b}{D_b} + A_b + M_{\text{shared}} + M_{\text{runtime}}. \quad (8)$$

FSDP/ZeRO, tensor/context parallelism, and activation checkpointing remain applicable inside an oversized block. With disjoint hardware, local critical time approaches $\max_b T_b$ rather than $\sum_b T_b$; aggregate work is not thereby reduced.

Teacher materialization is not free. Let T_{cache} be cache generation, T_{asm} assembly/evaluation, and K the number of runs reusing a cache:

$$T_{\text{ZG, total}} = T_{\text{cache}} + \max_b T_b + T_{\text{asm}}, \quad (9)$$

$$\bar{T}_{\text{ZG}}(K) = \frac{T_{\text{cache}}}{K} + \max_b T_b + T_{\text{asm}}. \quad (10)$$

If $T_{\text{local}} = \max_b T_b + T_{\text{asm}} < T_{\text{base}}$, reuse breaks even only when

$$K > \frac{T_{\text{cache}}}{T_{\text{base}} - T_{\text{local}}}. \quad (11)$$

The experiments report local critical paths, cache size, and accelerator-hours separately; they do not establish favorable first-use end-to-end economics against FSDP or ZeRO-3.

IV. EXPERIMENTAL METHOD

A. Unanchored GPT-2 Large Diagnostic

The preregistered study uses revision-pinned GPT-2 Large (774,030,080 parameters) and a SHA-256-pinned ten-million-token GPT-2 encoding of FineWeb-Edu. Each arm consumes 1,000,192 unique tokens at sequence length 64. The control is four-rank DDP with activation checkpointing. The treatment maps four nine-layer blocks to four disjoint L4 GPUs. Both arms execute the same steps, unique tokens, and aggregate residual-layer-token traversals; duplicated local vocabulary-interface work remains in the accounting.

The reported evidence is one completed matched seed (20260911). A planned second seed and $B = 2$ ablation were stopped for budget and excluded. Each local block ran 3,907 AdamW steps at 5×10^{-5} with BF16 compute. The objective used equi-probability noise intervals, 5% overlap, a weak 10^{-5} initialization anchor, and no teacher or context-margin term. Quality uses the same 8,192 held-out tokens, fixed corruptions,

contexts, and frozen source scorer. Thresholds were locked before the completed arms.

B. TinyLlama Materialized-Supervision Conversion

The follow-up uses revision-pinned TinyLlama base (1,100,048,384 parameters), TinyLlama Chat as frozen teacher, and UltraChat 200k. We deterministically select 8,192 first-turn exchanges for training and 512 disjoint validation examples at sequence length 256. The 22 decoder layers are partitioned $[0, 6)$, $[6, 12)$, $[12, 17)$, $[17, 22)$ across four L4s. The control is four-rank DDP. Each treatment worker trains one base-initialized partition against cached noise-conditioned teacher boundaries, a clean anchor, and final-block assistant cross entropy. Teacher embedding, RMSNorm, and LM head are frozen.

The locked ladder uses 128-step pilots, a 2,048-step primary (2,097,152 tokens), and a 256-step confirmation seed. The unanchored pilot cannot escalate when any learning, context, generation, or held-out gate fails. The cached replay must reproduce the earlier online-reference checkpoint hashes and assembled metrics. Five BF16 boundary tensors occupy 40 GiB of persistent storage; cache creation is outside timed local updates.

C. Six-Arm FSDP2 and Objective-Isolation Matrix

We strengthen that comparison with six arms under the same pinned base, teacher, dataset, 256-token shape, deterministic data stream, AdamW settings, BF16 compute, and 2,097,152-token primary budget. Arm A replaces DDP with global assistant-CE SFT using four-way FSDP2 and activation checkpointing. Arm B is the unanchored four-block objective and is pilot-gated. Arm C trains the four partitions against cached teacher boundaries only at $\sigma = 0$. Arm D is materialized ZEROGRAPH with cached boundaries and log-uniform $\sigma \in [0.005, 0.10]$. Arm E recomputes D's targets from the teacher online. Arm F changes D to $B = 2$, with each independent block sharded over a disjoint two-L4 FSDP2 group.

Thus A versus D is the strong conventional systems comparison; B tests whether unanchored local denoising preserves language semantics; C versus D isolates noise conditioning beyond hidden-state regression; D versus E isolates materialization rather than objective; and D versus F tests composition with state sharding. The initial 128-step gate and its post-pilot,

append-only 256-step correction are disclosed in Appendix J. Primaries escalate only after the corrected learning and unchanged assembled-quality gates pass. Evaluation includes held-out CE and accuracy, context ablation, fixed-prompt generation, teacher-scored generations, and boundary mixtures $\alpha h^T + (1 - \alpha)h^{\text{assembled}}$ for $\alpha \in \{1, .75, .5, .25, 0\}$. A second fixed seed confirms D. All timings separate cache construction, local critical compute, evaluation, and checkpoint I/O; FSDP2 tensor payload is labeled algorithmic rather than fabric traffic.

V. RESULTS

A. Unanchored Blocks Optimize but Lose Semantics

Table II reports a $2.41\times$ peak-HBM reduction and $6.30\times$ shorter local critical path. All four local loss windows improved, and assembled denoising MSE fell from 100.0182 to 0.5638. Yet held-out CE rose from 3.6342 to 10.4811, and token accuracy fell from 34.89% to 5.74%.

Every low-noise context effect was negative (-1.02% , -1.03% , -1.22% , and -2.20%): bypassing context slightly improved denoising. The source model scored ordinary continuations at CE 2.1558 and ZEROGRAPH continuations at 7.5821. The combined gate status is fail. Thus the blocks learned the corrupted target representation without learning a compositionally useful causal transformation.

Strict ZEROGRAPH reported zero inter-block gradient bytes. The DDP ring-allreduce payload estimate was 72.58 TB; this is algorithmic, not fabric-counter traffic. Excluding checkpoint I/O, DDP consumed 6.155 accelerator-hours and ZEROGRAPH consumed 0.974. Duplicated local interfaces increased final checkpoint storage from 9.29 GB to 17.61 GB. Active HBM reduction is not storage reduction.

B. Materialized Supervision Restores Composition

The unanchored TinyLlama pilot failed generation coherence and did not escalate. Every anchored pilot block learned, and the assembled model passed all gates.

Anchored CE was 1.3096, 0.0305 nat above ordinary SFT and 0.0160 below the frozen teacher. The context-ablation delta was 0.8279 nat, with four of four positive prompt deltas; teacher-scored generation CE was 0.8892 versus 0.5967 for teacher generations. The 256-step confirmation seed passed every gate at CE 1.3574.

The cached replay reproduced 4/4 online checkpoint hashes and all held-out metrics; maximum assembled teacher-boundary MSE was 0.0026. Peak HBM fell by $4.62\times$, and the local critical path was $4.90\times$ shorter, with 0 inter-block gradient bytes. The complete campaign cost \$3.39 under its locked budget. These measurements establish one conversion recipe on one architecture, not a universal speed or quality law.

C. Strong FSDP2 and Objective Controls

The six-arm matrix sharpens that result. Table V reports the locked primaries; B is shown separately because its failed pilot could not escalate. Every primary consumed 2,097,152 tokens. C, cached boundary regression at $\sigma = 0$, achieved the best CE (1.3059) and passed every context and generation

gate. D improved teacher-forced CE to 1.3173 and had a large positive context-ablation effect (7.3563 nats), yet its free-running outputs collapsed: maximum token share was 0.664 and repeated-bigram fraction was 0.675. Its primary gate therefore failed, and the locked stop rule barred the confirmation seed. F failed the same coherence gate.

This isolates the scientific contribution more narrowly. C minus D held-out CE was -0.0114 nat, and only C generated coherently; under this recipe, diffusion noise did not improve composition beyond FitNets-style boundary regression. The unanchored B pilot likewise failed coherence at CE 1.5554. Materialized boundary supervision itself remains the positive result, while the tested language DiffusionBlocks objective is a negative result.

Against the strong FSDP2 control, C's local path was 536.7 s and its cache-inclusive first use was 873.8 s, $4.079\times$ shorter than A's 3564.4 s compute path in this allocation. Peak HBM was nearly equal (A/C: $1.011\times$), so this experiment does *not* establish a memory advantage over four-way FSDP2. A's algorithmic collective payload estimate was 18.44 TiB/rank; strict C/D/E and inter-block F traffic were zero. These are tensor-payload estimates, not measured fabric counters, and the comparison is conversion/distillation versus ordinary SFT rather than objective parity.

D and E produced all four blocks in bitwise-equal canonical parameter states and identical CE, proving that caching changed information placement rather than the executed objective. Caching reduced peak HBM by $1.400\times$ and made E's local path $1.090\times$ longer, but the 40-GiB cache cost 337.0 s to create. It amortized below online recomputation only at $K = 8$ among $K \in \{1, 2, 4, 8\}$. F demonstrates that independent objectives compose mechanically with inner FSDP2, but its $4.112\times$ longer critical path shows that this two-GPU-per-block L4 configuration was inefficient. At the final block, fully assembled predecessor states increased D's boundary MSE from 0.006398 to 0.017747 ($2.77\times$), quantifying train-inference boundary shift. The complete locked campaign cost \$8.09.

D. Evidence-First Strengthening Campaign

We next held the clean teacher-boundary signal fixed and changed only where it was produced. Recomputed online boundaries and the persistent BF16 cache agreed bitwise on all 12 tested boundaries (maximum MSE zero). Across 16 paired short trials, cached jobs averaged 99.13 s after materialization versus 116.39 s online. The 72.0-GiB cache cost 618.3 s to create, however, so cache-inclusive amortized times at $K = 1, 2, 4, 8, 16$ were 724.0, 412.9, 256.6, 176.5, and 137.8 s/run. Online averaged 116.4 s at $K = 16$; no measured break-even occurred. Materialization shortened recurring target acquisition and changed the dependency graph, but did not establish an end-to-end wall-time advantage in this reuse range.

The 16-pair cached/online block experiment is the strict objective-matched test; Table VI provides broader systems references. Five-task zero-shot mean accuracy was 0.646 for ZEROGRAPH and 0.628 for FSDP2. The clean-boundary block-count sweep passed all gates: held-out CE was 1.2937, 1.2779,

TABLE II
COMPLETED MATCHED GPT-2 LARGE SEED ON FOUR L4S. COMPUTE EXCLUDES CHECKPOINT I/O; ZEROGRAPH TIME IS THE SLOWEST CONCURRENT BLOCK.

Arm	GPUs	B	Peak HBM	Compute	Held-out CE	Accuracy
DDP + checkpointing	4	1	14.71 GiB	5539.30 s	3.6342	34.89%
ZEROGRAPH unanchored	4	4	6.11 GiB	879.06 s	10.4811	5.74%

TABLE III
PREDECLARED GPT-2 LARGE GATES.

Gate	Threshold	Outcome
Peak HBM	$\geq 1.5\times$ reduction	pass
Critical compute	$\geq 1.1\times$ shorter	pass
Local learning	all blocks improve	pass
Assembled denoising	$\geq 20\%$ MSE reduction	pass
Low- σ context	every block $> 1\%$	fail
Generation retention	$\Delta\text{CE} \leq 2.0$	fail

TABLE IV
TINYLLAMA PRIMARY ON FOUR L4S. CRITICAL COMPUTE EXCLUDES CACHE CONSTRUCTION, LOADING, AND CHECKPOINT I/O.

Arm	Peak HBM	Critical compute	Held-out CE
Frozen base	—	—	1.6813
Frozen Chat teacher	—	—	1.3256
Ordinary DDP SFT	12.32 GiB	1936.41 s	1.2791
Materialized ZEROGRAPH	2.67 GiB	395.20 s	1.3096

and 1.2742 for $B = 2, 4, 8$. Interpolating teacher and assembled predecessor states increased late-boundary error as assembled-state weight grew, directly connecting representation drift to boundary mismatch, although CE remained stable in the measured range. This does not rescue the separate noisy objective, whose free-running coherence failure replicated.

Qwen2.5-3B (3,085,938,688 parameters) then trained as four disjoint jobs for 2,048 updates each. Assembled CE was 1.3002 versus 1.2309 for the base and 1.3600 for the teacher; context delta was +0.509 nat with four positive prompt deltas. All gates passed, peak allocated HBM was 13.59 GiB, and inter-block gradient traffic was zero. This is real 3B conversion/distillation, not from-scratch pretraining. Separately, one job was killed by SIGTERM halfway and resumed with optimizer, RNG, and sample order; its final state was bitwise identical to an uninterrupted deterministic control (digest prefix 238f1f1d7e3e). The other three jobs required no retraining or rollback. The verified nine-phase campaign cost \$33.01.

VI. RELATED WORK

Checkpointing and dynamic rematerialization preserve the ordinary objective while reducing activation retention [1], [10]; ZeRO, FSDP, Megatron, and pipeline parallelism redistribute state or execution while retaining global credit [2]–[5]. EXACTGRAPH is our control in this objective-preserving family.

Synthetic gradients and greedy layerwise learning reduce locking through local or predicted signals [11], [12]. ZEROGRAPH instead uses the DiffusionBlocks noise-local objective [6], grounded in score/SDE and EDM denoising [7], [8].

TABLE V
TINYLLAMA SIX-ARM COMPARISON ON FOUR L4S. TIME IS THE LOCAL CRITICAL COMPUTE PATH AND EXCLUDES CACHE CONSTRUCTION, EVALUATION, AND CHECKPOINT I/O.

Arm	Learning signal	Peak HBM	Time	CE	Generation
A	global CE, FSDP2	5.18 GiB	3564.4 s	1.4643	pass
C	cached boundary, $\sigma = 0$	5.13 GiB	536.7 s	1.3059	pass
D	cached boundary + noise	5.13 GiB	531.3 s	1.3173	fail
E	online boundary + noise	7.18 GiB	578.9 s	1.3173	fail
F	D, $B = 2$, inner FSDP2	4.75 GiB	2184.6 s	1.3366	fail

TABLE VI
QWEN2.5-1.5B ACCOUNTING ON FOUR L4S. FIRST USE INCLUDES CACHE CONSTRUCTION WHERE APPLICABLE; REUSE DOES NOT. SFT AND DISTILLATION OBJECTIVES ARE NOT TREATED AS OBJECTIVE MATCHED.

Arm	First use	Reuse	Peak GiB	CE
FSDP2 SFT	5034.3	5034.3	7.47	1.4377
ZeRO-3 SFT	3496.5	3496.5	11.52	1.4399
Global boundary distill	5275.3	4657.0	10.71	1.3093
ZEROGRAPH cached boundary	1169.5	551.1	7.45	1.2779

BlockTrain independently studies local text objectives and block-level transport [13].

Knowledge distillation transfers behavior from teachers to students [14]; FitNets and later Transformer distillation methods match intermediate representations [15]–[17]. We do not claim teacher hidden-state targets as new. Our question is whether an authenticated, reusable boundary trajectory can serve as the systems interface that removes the teacher and neighboring trainable blocks from each local optimization loop. The failure–repair sequence makes this distinction measurable: unanchored jobs lose compatible semantics; materialized boundaries restore a reference trajectory while preserving independent optimization.

VII. LIMITATIONS AND REPRODUCIBILITY

The strongest positive result is teacher-anchored conversion, not from-scratch pretraining. It covers one architecture, one 2.10M-token primary, and a short confirmation seed. Teacher scoring and held-out CE do not replace broad downstream or human evaluation. The executed recipe combines boundary targets, clean anchoring, final-block assistant CE, and frozen interfaces; it establishes sufficiency, not the marginal effect of each term.

Training uses teacher predecessor states while inference uses learned states. The low assembled boundary MSE is encouraging but does not establish robustness under accumulated boundary shift. The five cached tensors occupy 40 GiB, and

cache construction is excluded from the local critical-path comparison. Equation (10) was evaluated through $K = 16$; cached execution remained slower after amortizing construction, so no break-even is claimed. The strengthening campaign adds ZeRO-3, three 1.5B seeds, and a real 3B replication, but still covers only Qwen2.5-family models, short adaptation budgets, one four-L4 provider topology, and automatic rather than human language evaluation.

The six-arm result also narrows the algorithmic claim. Cached $\sigma = 0$ boundary regression passed generation gates, while the executed noise-conditioned D/E variants and hierarchical F variant did not. Their strong teacher-forced CE and context-ablation scores therefore cannot be interpreted as usable language quality. The preregistered stop rule barred D’s confirmation after its primary failure. C’s positive result is hidden-state distillation used as a systems interface, not evidence that diffusion is necessary, and its comparison with ordinary SFT is not objective parity. F proves mechanical composition with FSDP2 but was slower on the tested L4 allocation; no universal sharding law is claimed.

The GPT-2 Large study has one completed seed; the planned replication and block count ablation were not completed. Its negative outcome is diagnostic, not a proof against all unanchored objectives. All numerical macros are generated from retained evidence. The verifier checks 50 claims, cross-file invariants, checkpoint identities, and complete source digests. The supplementary appendix contains the ExactGraph control, 4.027B systems fixture, smaller studies, expanded protocols, and traceability anchors.

VIII. CONCLUSION

Global backpropagation couples depth because upstream credit depends on downstream information. Rematerialization can change storage and execution but cannot make the ordinary objective depth-independent. ZEROGRAPH changes the contract. Unanchored GPT-2 Large blocks met the mechanical contract and failed the semantic one. Cached TinyLlama teacher boundaries supplied a reference trajectory before optimization, after which each block trained as an independent, resumable job with zero strict-mode inter-block gradient traffic.

The six-arm study identifies the boundary of that positive result. Plain cached boundary regression passed every language gate and had a substantially shorter cache-inclusive path than four-way FSDP2 in this allocation. Adding the tested DiffusionBlocks noise intervals improved neither CE nor composition and caused free-running collapse; cached and online forms were bitwise identical, so this failure belongs to the objective rather than materialization. Hierarchical FSDP2 was mechanically valid but inefficient on the tested topology.

ZEROGRAPH is therefore best interpreted as an information-placement primitive for pretrained-model conversion, not a validated diffusion replacement for language pretraining. The evidence-first campaign confirms this on Qwen2.5 at 1.5B and 3B scale against FSDP2 and ZeRO-3, with three seeds, block-count and boundary sweeps, cache-inclusive accounting, and bitwise independent restart. Its most important negative result is

TABLE VII
PER-BLOCK UNANCHORED EVIDENCE. LOSS IS THE FIRST-100 TO LAST-100 MEAN; $\Delta_{ctx} > 0$ INDICATES USEFUL CAUSAL CONTEXT.

b	Layers	Noise interval	Loss	Δ_{ctx}
0	1–9	[1.577, 10]	11.753 $\rightarrow$ 10.402	–1.02%
1	10–18	[0.782, 1.577]	113.567 $\rightarrow$ 10.592	–1.03%
2	19–27	[0.421, 0.782]	14.850 $\rightarrow$ 10.796	–1.22%
3	28–36	[0.200, 0.421]	28.676 $\rightarrow$ 10.831	–2.20%

equally clear: even at $K = 16$, cache amortization did not beat objective-matched online target generation. The supported claim is that ZeroGraph converts cross-depth semantic dependence from an online optimization dependency into a persistent data dependency, yielding independently schedulable and recoverable jobs; its economics depend on reuse, storage, topology, and target-generation cost.

APPENDIX A EXPANDED GPT-2 LARGE PROTOCOL

The source checkpoint was divided into four contiguous nine-layer blocks. Each job contained 371.3M local parameters, of which 307.1M were copied from the checkpoint and 242.7M were trainable under the local objective; the frozen vocabulary/target interface was duplicated by construction. Noise used $\log \sigma \sim \mathcal{N}(-0.5, 1.2^2)$ truncated to $[0.2, 10]$, equal-mass intervals, and 5% boundary overlap. Each block ran 3,907 AdamW steps at 5×10^{-5} , batch four, sequence length 64, BF16 compute, gradient-norm cap 1.0, and a 10^{-5} initialization anchor. No teacher or context-margin term was active. Assembly used two Heun steps per interval and denoised to zero.

The baseline and treatment consumed 1,000,192 unique tokens. Five-hundred step checkpoints bind optimizer, data stream, RNG, and cumulative compute time so a platform retry cannot shorten the denominator. NVML sampling retained utilization and power; framework telemetry recorded peak allocated/reserved HBM. The DDP communication number is an algorithmic ring-allreduce payload estimate, not a fabric counter. The provider’s containerized topology query was unavailable, so no physical-fabric claim is made.

The baseline’s success at the same token budget rules out insufficient tokens as a complete explanation, although one million tokens is too small for a general convergence claim. The evidence is consistent with three mechanisms: the corrupted target provides a shortcut around context, independent output spaces drift, and the weak anchor permits rapid departure from the pretrained function. More optimization of the same objective is not an established cure.

APPENDIX B EXPANDED TINYLLAMA PROTOCOL AND ACCOUNTING

The exact seeded stream materialized five BF16 boundaries occupying 40 GiB. The teacher body was absent during timed local updates. Primary evidence required held-out CE within 0.5 nat of the teacher, teacher-scored generation within 1.0 nat, coherent fixed-prompt generations, and positive context

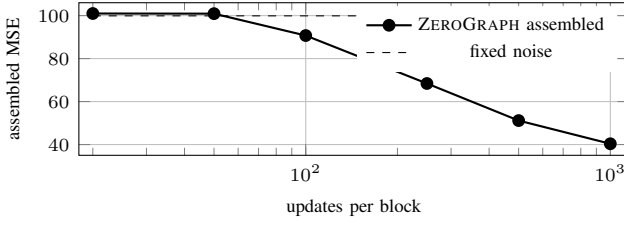


Fig. 2. Held-out trajectory for the synthetic 4.027B representation fixture.

ablation for at least three prompts. The cached replay had to match an earlier online-reference execution exactly.

The successful recipe is a sufficient repair package, not a component-level causal ablation. Relative to the failed unanchored pilot, it jointly adds cached teacher boundaries, a clean initialization anchor, final-block assistant CE, and frozen teacher-side interfaces. The evidence does not isolate the marginal contribution of each term.

First-use wall time must include teacher execution, boundary writes, local training, reads, and assembly. The cache can be reused across hyperparameter trials or adaptations, but reuse amortizes rather than erases its cost. The reported campaign cost was \$3.39; the stopped orchestration ledger contains 13 entries and remained below its locked ceiling.

A. Boundary-Shift Evaluation Needed

Training worker b on h_{b-1}^T creates teacher forcing at a depth boundary. The assembled model instead supplies $\hat{h}_{b-1}$. The observed maximum teacher-boundary MSE was 0.0026, but a definitive robustness study should train or evaluate with

$$\tilde{h}_{b-1} = \alpha h_{b-1}^T + (1 - \alpha) \hat{h}_{b-1} + \eta, \quad (12)$$

where α is swept from one to zero and η is calibrated boundary noise. Reporting CE and boundary error across this sweep would distinguish a stable materialized interface from brittle teacher forcing.

APPENDIX C

4.027B SYSTEMS-SCALE FIXTURE

The fixture is a 20-layer BF16 causal representation Transformer, width 4096, 32 heads, sequence length eight, and 4,026,859,520 parameters. Compilation produced ten authenticated 402,702,336-parameter blocks. Targets are a deterministic nonlinear function of context over 32 fixed synthetic examples; eight are held out. Every block received 1,000 SGD updates. This validates ownership, working-set, scheduling, and assembly mechanics—not meaningful 4B-scale language learning.

The ordinary 4B BF16 SGD baseline peaked at 16.134 GB; one resumed local worker peaked at 1.637 GB. Serialized state was 8.054 GB globally and 805.4 MB per block. Full assembled evaluation peaked at 8.069 GB; independent training does not shrink assembled inference.

The ten-stream schedule was 3.84% faster than one stream and produced identical final tensor digests. Peak allocation increased 1.952 GB. Nsight recorded 8,150 kernels on ten

streams; 27.515 ms (2.30% of kernel-active union) had concurrency two. No NCCL, c10d, all-reduce, all-gather, or reduce-scatter kernel appeared. This proves concurrency but not a general multi-GPU speed law.

APPENDIX D

EXACTGRAPH OBJECTIVE-PRESERVING CONTROL

EXACTGRAPH uses the same structural partitions but preserves the ordinary loss. It rematerializes one partition at a time and propagates exact boundary VJPs. Certification compares initial state, output, separators, scalar loss, gradients, optimizer state, post-step parameters, buffers and aliases, inputs, and RNG bytes. Unsupported capture, mutation, nondeterminism, sparse gradients, or guard drift rejects before the optimized path runs.

The fixture is a five-layer BF16 causal residual Transformer of width 4096, 32 heads, sequence length eight, and 1,006,714,880 parameters. One complete optimizer step passed output, loss, 40 gradient tensors (2.013 GB), optimizer state, parameters, inputs, and RNG. Ordinary forward retained 77 non-state storages (5,510,904 bytes); rematerialized forward retained seven (589,824 bytes). Peak CUDA increments were unchanged because parameters and gradients dominated. The one-step timing is a mechanics measurement, not stable throughput.

An optional transport codec predicts a boundary adjoint and losslessly encodes the XOR residual, falling back to dense bytes. Eight observed adjoints reconstructed bitwise; the second warm step used 117,580 wire bytes versus 262,144 dense bytes. This changes the representation of an exact dependency, not the dependency itself.

APPENDIX E

SMALLER REAL-DATA CHECKS

An 11.1M-parameter conditional model trained on 4,096 MNIST examples achieved assembled MSE 0.08575 versus 1.10389 for noise and 0.01309 for the end-to-end source. A pretrained 81.9M-parameter DistilGPT-2 conversion trained two local blocks for 5,000 steps. Its preregistered WikiText gate passed: teacher-scored NLL 7.684, maximum token share 0.0983, mean unique-token fraction 0.7856, and normalized entropy 0.3571. Teacher NLL on real validation text was 5.815, exposing a material quality gap rather than parity.

APPENDIX F

TRACEABILITY AND CLAIM BOUNDARIES

The verifier checks 69 claims against retained JSON evidence and recomputes the generated result macros. Key SHA-256 prefixes are: EXACTGRAPH certificate 734c610c9c9; 4B report 7bfcf4f3071; concurrency manifest 4051cf9a4dd; TinyLlama orchestration 464db1743aee; primary evaluation 27253044ded6; and confirmation evaluation c635b6afcfb3. The six-arm verified-summary prefix is 28be510ee595, its independent verification is 6c98ee71d2cf, and primary evaluation is 571d94b79cb9. Complete digests, not prefixes, are verified. The final nine-phase campaign summary has SHA-256 prefix

TABLE VIII
SECONDARY SYSTEMS EVIDENCE. RETAINED FORWARD BYTES ARE NOT TOTAL HBM; THE 4B LOCAL PEAK IS ONE WORKER, NOT ASSEMBLED INFERENCE.

Experiment	Metric	Control	Method	Interpretation
EXACTGRAPH, 1.007B	Retained non-state bytes	5,510,904	589,824	$9.343\times$ less
	Step time (ms)	59.45	70.06	$1.178\times$ time
	Peak CUDA increment (GB)	2.014	2.014	state dominated
	Certified optimizer steps	—	1	bitwise pass
ZEROGRAPH, 4.027B	Peak allocated bytes	16,133,899,776	1,637,239,296	$9.854\times$ less
	Assembled MSE	99.8916	40.3645	59.6% lower
	Updates/block; aggregate	—	1,000; 10,000	optimization
	Inter-block gradient bytes	—	0	strict mode
ZEROGRAPH, one GB10	Wall time (s)	5.4339	5.2327	$1.038\times$
	Resident peak bytes	16,156,805,120	18,108,401,664	12.1% higher
	Final block digests	—	10/10 exact	invariant
	Kernel overlap	0	27.52 ms	max concurrency 2

702bd8265fab; its status is verified, all nine phases are complete, and its failure list is empty. The campaign generator fails closed on those invariants and reproduces both the LaTeX macros and ordered Markdown report from that summary.

For the GPT-2 Large pair, the checkpoint revision prefix is 32b71b12589c, model digest 5f47f3e12f91, token digest f0d18873b700, evaluation-code digest 8075c272042d, and DDP checkpoint digest ab5c7590d83a. Local report prefixes are 8d0446c1, b72a9874, ae454b56, and 3bce6d70. The root preregistration and 13 hash-chained operational addenda disclose canceled engineering attempts and the budget-stopped partial seed; only the completed matched pair contributes results.

APPENDIX G SUBMISSION CHECKLIST

- The main claim is conversion/distillation, not from-scratch pretraining.
- Local critical-path time excludes cache construction; reported first-use time adds the measured cache pass and excludes common evaluation/checkpoint I/O.
- Zero inter-block bytes applies only to frozen/private-interface strict mode.
- The GPT-2 result is one completed negative seed; the confirmation seed for TinyLlama is shorter than the primary.
- The six-arm D confirmation was barred by its preregistered primary coherence failure; C is the positive boundary-regression control.
- The 4B experiment is synthetic systems evidence, not 4B language quality.
- The 1B ExactGraph reduction is retained forward storage, not total HBM.
- The new comparison includes FSDP2, boundary-shift tests, and cache accounting; ZeRO-3, a second architecture, and broader downstream/human evaluation remain required.

APPENDIX H REPRODUCTION AND ARTIFACT MAP

From the repository root, the complete manuscript check is:

TABLE IX
PRIMARY REPOSITORY EVIDENCE LOCATIONS.

Evidence	Repository path
Claim table	papers/zerograph_ieee/claims.json
GPT-2 protocol	experiments/modal_l4_poc/
GPT-2 reports	artifacts/modal_l4_stopped_seed1/
TinyLlama protocol	experiments/modal_l4_tinyllama/
TinyLlama reports	artifacts/modal_l4_tinyllama/
Six-arm reports	artifacts/modal_l4_six_arm/
4B report	artifacts/independent_4b_training/
ExactGraph control	artifacts/exact_1b_cuda_v0_17_0_final/

make ieee-paper

This command regenerates the evidence verification report, compiles the main paper and supplement with the vendored IEEE class, and rejects a main paper over nine pages. The underlying fail-closed checks can be invoked directly:

```
cd papers/zerograph_ieee
python3 generate_l4_results.py --check
python3 generate_tinyllama_results.py \
  --check
python3 generate_six_arm_results.py \
  --check
python3 verify_evidence.py \
  --output evidence-verification.json
```

The manuscript fails when any generated result file is absent; a missing evidence input cannot silently produce a shorter paper with different claims.

APPENDIX I EVIDENCE-FIRST STRENGTHENING CAMPAIGN

The nine-phase protocol used revision-pinned Qwen2.5-1.5B checkpoints, UltraChat, length 256, 2,048 updates, and four L4s. It compared FSDP2/ZeRO-3 SFT, connected distillation, online boundaries, and cached ZEROGRAPH. It covered three seeds, $B \in \{2, 4, 8\}$, five interpolation weights, noise stress, five downstream tasks, 16 reuse pairs, INT8 compression, and failure/restart.

The objective-matched cached/online jobs held teacher, examples, targets, optimizer, updates, evaluation, and allocation

fixed. Twelve cache boundaries matched bitwise. Deterministic kernels were reserved for restart certification.

At 3B, fixed base/teacher revisions were partitioned $[0, 9)$, $[9, 18)$, $[18, 27)$, $[27, 36)$; each 693.7M-parameter job consumed 524,288 tokens. All local losses fell and assembly passed CE, context, and coherence gates. The claim remains pretrained conversion/distillation.

The lifecycle test sent SIGTERM to block 1 at step 256, restored it to step 512, and compared it with an uninterrupted control. Canonical final tensor digests matched bitwise; unaffected jobs were not rerun. Two failed engineering attempts—GPU-mapped RNG state and initially nondeterministic kernels—remain in the append-only ledger rather than being erased.

APPENDIX J

SIX-ARM FSDP2 COMPARISON PROTOCOL

The A–F comparison uses the same revision-pinned TinyLlama base, Chat teacher, UltraChat split, 256-token shape, and deterministic four-example stream as the conversion study. Each primary arm receives 2,048 updates (2,097,152 example tokens). Owned parameters and AdamW state are FP32; forward/backward compute is BF16, and FSDP2 reductions are FP32. Optimizer settings are AdamW with learning rate 2×10^{-5} , $\beta = (0.9, 0.95)$, zero weight decay, and gradient-norm cap 1.0.

Arm A is global assistant-only cross entropy with activation checkpointing and four-way FSDP2. Arm B assigns the unanchored local denoising objective to four independent jobs. Arm C uses the cached teacher trajectory at $\sigma = 0$, providing a hidden-state-regression control. Arm D uses the same cache with log-uniform $\sigma \in [0.005, 0.10]$. Arm E recomputes the identical D targets online. Arm F changes D to two depth partitions, each sharded over a disjoint two-GPU FSDP2 group. Thus C versus D isolates noise conditioning, D versus E isolates materialization, and D versus F tests composition with state sharding.

The first 128-step pilot exposed a declared-gate mismatch: the final block’s mixed scalar loss includes batch-variable assistant CE, despite the gate’s intended use as a denoising-learning check. The original failure is retained. Before further allocation, an append-only addendum locked a 256-step diagnostic and required every block’s denoising component to decrease while all previously locked held-out, context, and coherence gates remained unchanged. Arm B failed local/coherence gates and could not escalate. This amendment is post-pilot and must not be described as part of the original preregistration.

Boundary robustness mixes teacher and assembled predecessor states at fractions $\alpha \in \{1, 0.75, 0.5, 0.25, 0\}$ and reports teacher-target MSE for every block. Communication accounting separates strict inter-block gradient bytes from algorithmic intra-block FSDP2 tensor payload. Materialization economics reports the cache pass, local critical path, first use, and amortized reuse at $K \in \{1, 2, 4, 8\}$. The incremental Modal ledger has a hard \$25 ceiling.

All A/C/D/E/F primaries completed without OOM, and the final ledger was \$8.09. C passed every assembled gate. D and

E were bitwise identical but failed coherence; F also failed coherence. The preregistered rule conditioned a second D seed on the primary quality gate, so it was not launched. This is a stopped negative result, not missing-at-random replication. The compact bundle retains both preregistrations, every gate report, per-arm telemetry summaries, the cost ledger, exact executed-source hashes, and an independent verification record.

REFERENCES

- [1] T. Chen, B. Xu, C. Zhang, and C. Guestrin, “Training deep nets with sublinear memory cost,” in *Proc. NIPS Workshop on Memory-Efficient Machine Learning*, 2016.
- [2] S. Rajbhandari, J. Rasley, O. Ruwase, and Y. He, “ZeRO: Memory optimizations toward training trillion parameter models,” in *Proc. SC*, 2020, pp. 1–16.
- [3] Y. Zhao, A. Gu, R. Varma, L. Luo, C.-C. Huang, M. Xu, L. Wright, H. Shojanazeri, M. Ott, S. Shleifer, A. Desmaison, C. Balioglu, B. Nguyen, G. Chauhan, Y. Hao, and S. Li, “PyTorch FSDP: Experiences on scaling fully sharded data parallel,” *Proc. VLDB Endowment*, vol. 16, no. 12, pp. 3848–3860, 2023.
- [4] M. Shoenberger, M. Patwary, R. Puri, P. LeGresley, J. Casper, and B. Catanzaro, “Megatron-lm: Training multi-billion parameter language models using model parallelism,” *arXiv preprint arXiv:1909.08053*, 2019.
- [5] Y. Huang, Y. Cheng, A. Bapna, O. Firat, M. X. Chen, D. Chen, H. Lee, J. Ngiam, Q. V. Le, Y. Wu, and Z. Chen, “GPipe: Efficient training of giant neural networks using pipeline parallelism,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [6] M. Shing, M. Koyama, and T. Akiba, “Diffusionblocks: Block-wise neural network training via diffusion interpretation,” *Proc. ICLR*, 2026, arXiv:2506.14202.
- [7] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, “Score-based generative modeling through stochastic differential equations,” in *Proc. ICLR*, 2021.
- [8] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 26 565–26 577.
- [9] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [10] M. Kirisame, S. Lyubomirsky, A. Haan, J. Brennan, M. He, J. Roesch, T. Chen, and Z. Tatlock, “Dynamic tensor rematerialization,” in *Proc. ICLR*, 2021.
- [11] M. Jaderberg, W. M. Czarnecki, S. Osindero, O. Vinyals, A. Graves, D. Silver, and K. Kavukcuoglu, “Decoupled neural interfaces using synthetic gradients,” in *Proc. ICML*, 2017, pp. 1627–1635.
- [12] E. Belilovsky, M. Eickenberg, and E. Oyallon, “Greedy layerwise learning can scale to imagenet,” in *Proc. AISTATS*, 2019, pp. 583–593.
- [13] P. Toth and D. Oprisa, “Decentralised ai training and inference with blocktrain,” *arXiv preprint arXiv:2606.24722*, 2026.
- [14] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015.
- [15] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio, “FitNets: Hints for thin deep nets,” in *Proc. ICLR*, 2015.
- [16] S. Sun, Y. Cheng, Z. Gan, and J. Liu, “Patient knowledge distillation for BERT model compression,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 4323–4332.
- [17] X. Jiao, Y. Yin, L. Shang, X. Jiang, X. Chen, L. Li, F. Wang, and Q. Liu, “TinyBERT: Distilling BERT for natural language understanding,” in *Findings of EMNLP*, 2020, pp. 4163–4174.